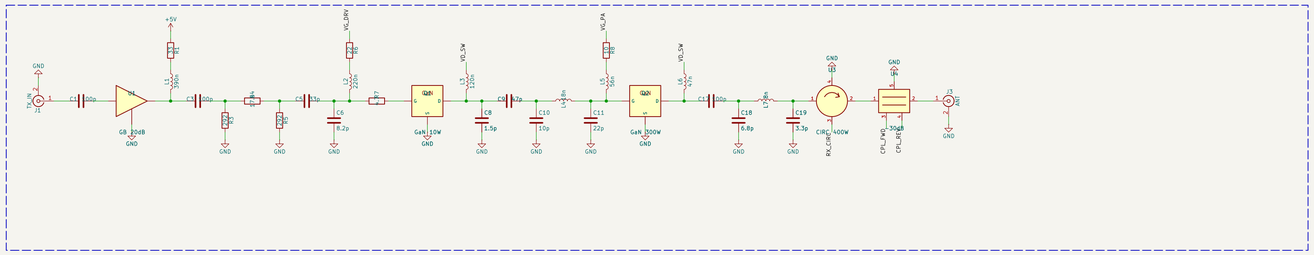
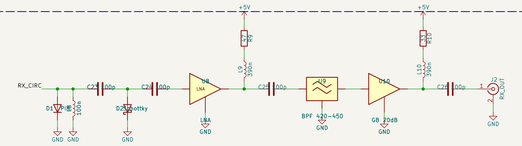
TX CHAIN — 0 dBm in, 300 W (+54.8 dBm) peak out, 20 % max duty



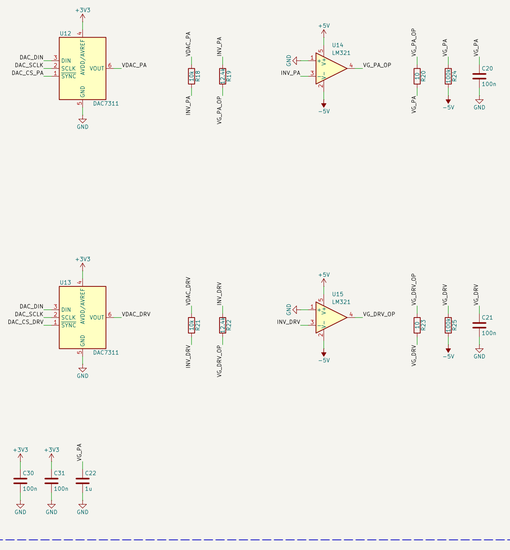
RX CHAIN — limiter protected, NF 1.1 dB, G 34 dB



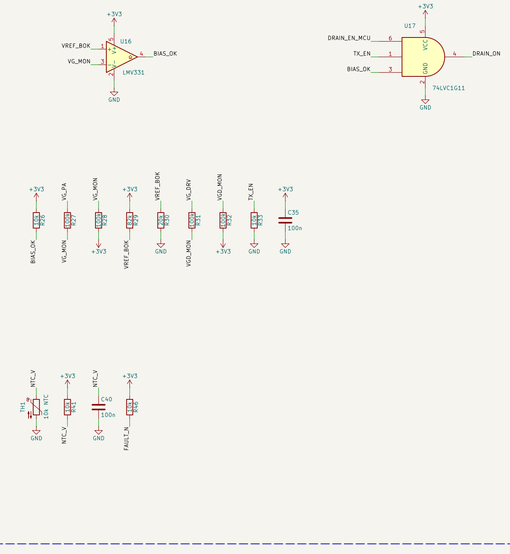
POWER MONITOR AND CALIBRATION PORT



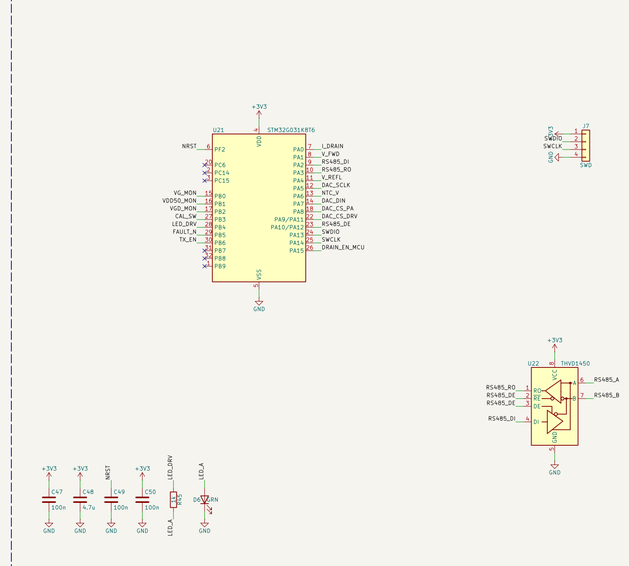
GATE BIAS — VG = −1.25 x VDAC (−1.0 … −4.1 V), fail-safe pinch-off



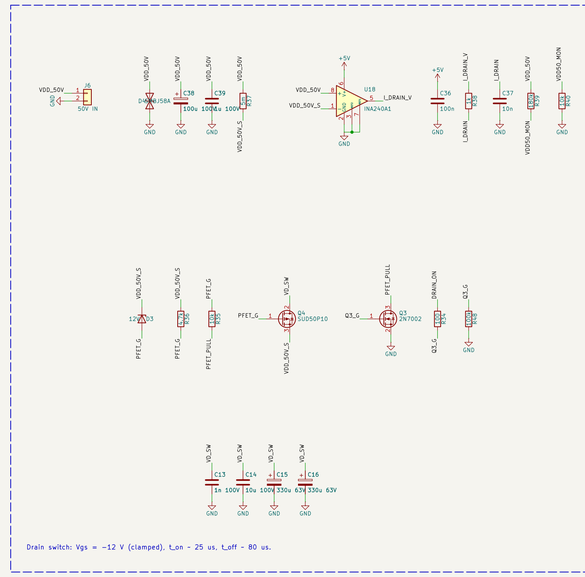
SEQUENCING INTERLOCK AND MONITORS



MODULE CONTROLLER — STM32G031, RS-485 to tile, SWD

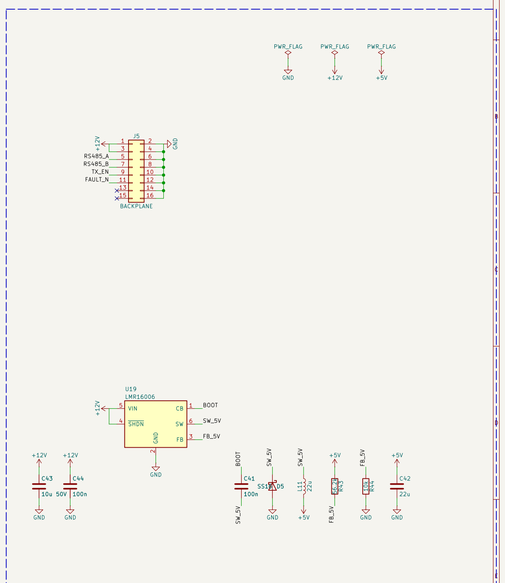


DRAIN SUPPLY — 50 V, 10 A peak, pulsed drain switch + current sense



Drain switch: Vgs = −12 V (clamped). t_on ≈ 25 us, t_off ≈ 80 us.

POWER — 12 V aux in, 5 V buck, 3.3 V LDO, −5 V inverter



MOUNTING / COLD PLATE



NOTES

1. Matching values are first-pass from transistor large-signal models; tune at 435 MHz on load-pull bench.
2. Q2 and U3 mount through board cutouts onto the copper cold plate (liquid cooled, 45 W avg per module).
3. RF lines: 50 ohm microstrip, 1.70 mm on RO4350B 0.762 mm (L1–L2). Output path >= 3 mm wide.
4. TX_EN is the only path that can turn the drain on; MCU and BIAS_OK can veto it in hardware.
5. Module power: 155 W avg RF per module x 2.587 per face; see radar_sizing.py for the system budget.

Generated by gen_schematic.py — first-pass design, not released for fabrication

SENTINEL-UHF reference design

Sheet: /

File: SENTINEL_TRM.kicad_sch

Title: SENTINEL-UHF T/R Module, 420–450 MHz, 300 W

Size: A1	Date: 2026-10-18	Rev: A
KiCad E.D.A. 10.0.3		Id: 1/1